Solving the Model, Step by Step

This document derives the solution of the model in the companion model section, one substitution at a time. It adds no new assumptions and no new results. Its purpose is to show exactly how the equilibrium conditions are reduced to a small system that can be solved, which objects have closed forms, and which ones must be computed. Propositions and theorems cited by name and number are those of the model section.

The derivation is written for the baseline: quantity competition, an upper-tier elasticity $\eta = 1$, intermediate inputs with shares $\nu_k \in [0, 1)$, and non-negative tariffs. Boxed notes say what changes under price competition and when $\eta > 1$, which the model section allows only without intermediate inputs.

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0 What Has to Be Solved

Primitives. The economy is described by: the sets of countries $\mathcal{N}$ and sectors $\mathcal{K}$; labor endowments L_n; preference weights β_k and elasticities $\sigma_k > \eta = 1$; intermediate shares ν_k and input–output weights $\omega_{k'k}$; plant productivities φ_i (which already contain the location, plant and group components and the head-office capability terms); the efficiency losses $\gamma_{h\ell}$; trade costs $d_{\ell n}$; tariffs $t_{\ell nk}$; fixed-cost parameters f_k and $f^d_{\ell n}$; the pool of potential groups with their potential plants $\mathcal{P}_g$; and ownership shares θ_{gn}. All of these are fixed before any decision is taken.

Unknowns. An equilibrium (Definition 2 of the model section) consists of wages w_n; sectoral price indices P_{nk}; market sizes E_{nk}; incomes X_n; active sets $T_{g,n} \subseteq \mathcal{P}_g$, one per group and destination of its sector; and the price $p_{i,n}$ and quantity $q_{i,n}$ of every active plant in every market it serves.

Strategy. There is no closed form for the equilibrium as a whole. What the model allows is a chain of substitutions in which every object becomes a function of two things only: the wage vector w and the configuration of active sets $T = \{T_{g,n}\}$. Each link of the chain is either a closed form or a problem with a unique solution that is easy to compute. The equilibrium is then a pair (w, T) that satisfies $N-1$ labor-market conditions and an entry condition in every market. The steps are:

Step	Solves for	Takes as given	How
1	demand functions	prices, spending	closed form
2	unit and delivered costs	wages, input prices	closed form
3	shares, markups, prices in one market	delivered costs, active plants	one scalar equation
4	active plants in one market	costs, fixed costs, spending	ladder and bisection
5	input prices	wages, active sets	contraction
6	market sizes and incomes	wages, active sets, prices	linear system
7	labor-market conditions	everything above	closed form
8	wages	active sets	$N-1$ equations
9	active sets	everything above	outer loop

Steps 1–3 and 7 hold at any wages. Steps 3–6 are nested: market shares do not depend on spending, so input prices can be found before market sizes, and market sizes and incomes solve a linear system once prices are known.

1 Step 1: Household Demand

Lower tier. Fix country n and sector k, and write $\sigma = \sigma_k$. For a given amount of the sectoral composite C_{nk}, the household minimizes spending $\sum_{i \in \mathcal{A}_{nk}} p_{i,n} q_{i,n}$ subject to $\left[\sum_i q_{i,n}^{(\sigma-1)/\sigma}\right]^{\sigma/(\sigma-1)} = C_{nk}$. With multiplier χ, the first-order condition for variety i is

$$p_{i,n} = \chi \, \frac{\partial C_{nk}}{\partial q_{i,n}} = \chi \, C_{nk}^{1/\sigma} q_{i,n}^{-1/\sigma}, \qquad \text{so} \qquad q_{i,n} = \left(\frac{p_{i,n}}{\chi}\right)^{-\sigma} C_{nk}.$$

Substituting into the constraint, $C_{nk} = \chi^{\sigma} C_{nk} [\sum_i p_{i,n}^{1-\sigma}]^{\sigma/(\sigma-1)}$, so $\chi = [\sum_i p_{i,n}^{1-\sigma}]^{1/(1-\sigma)} \equiv P_{nk}$. The multiplier is the price index, and minimized spending is $\sum_i p_{i,n} q_{i,n} = P_{nk} C_{nk}$. Hence

$$q_{i,n} = \Big(\frac{p_{i,n}}{P_{nk}}\Big)^{-\sigma} C_{nk} = p_{i,n}^{-\sigma} P_{nk}^{\sigma-1} \, (P_{nk} C_{nk}). \tag{1}$$

Upper tier. With $\eta = 1$ the upper tier is Cobb–Douglas, $U_n = \prod_k C_{nk}^{\beta_k}$, and maximizing it subject to $\sum_k P_{nk} C_{nk} = X_n$ gives the expenditure shares $P_{nk} C_{nk} = \beta_k X_n$. Final spending on sector k in n is therefore a fixed share of income,

$$E_{nk}^{F} = \beta_k X_n. \tag{2}$$

Total demand. Firms buy intermediate inputs as the same sectoral composites (Step 2), so they face the same price index and split their purchases across varieties by the same rule (1). Adding households and firms, and writing E_{nk} for total spending on sector k in n, demand for plant i's variety in n is

$$q_{i,n} = p_{i,n}^{-\sigma} P_{nk}^{\sigma-1} E_{nk}, \qquad P_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} p_{i,n}^{1-\sigma} \Big]^{1/(1-\sigma)}. \tag{3}$$

The revenue share of plant i in the market is $s_{i,n} \equiv p_{i,n} q_{i,n} / E_{nk} = p_{i,n}^{1-\sigma} / P_{nk}^{1-\sigma}$, and these shares add up to one.

When $\eta > 1$. Only without intermediates. The upper tier gives $P_{nk} C_{nk} = \beta_k (P_{nk}/P_n)^{1-\eta} X_n$ with $P_n = [\sum_k \beta_k P_{nk}^{1-\eta}]^{1/(1-\eta)}$, so final spending on a sector rises when its relative price falls. Everything below goes through with β_k replaced by $\beta_k (P_{nk}/P_n)^{1-\eta}$, which is known once prices are known (Step 5), so Step 6 is still a linear system in incomes.

2 Step 2: Costs

Unit cost. Plant i in sector k, located in ℓ, combines labor l with a bundle B of intermediate inputs, which in turn combines quantities $m_{k'}$ of the sectoral composites bought in ℓ:

$$y = \frac{\varphi_i}{\gamma_{h_i \ell}} \Big(\frac{l}{1-\nu_k}\Big)^{1-\nu_k} \Big(\frac{B}{\nu_k}\Big)^{\nu_k}, \qquad B = \prod_{k'} \Big(\frac{m_{k'}}{\omega_{k'k}}\Big)^{\omega_{k'k}}.$$

The constants in the denominators are the normalization under which the Cobb–Douglas cost function has no extra constant; the model section absorbs this constant into φ_i, which is equivalent. Minimizing $\sum_{k'} P_{\ell k'} m_{k'}$ for one unit of B gives the cost of the bundle, $P_{\ell k}^{\mathrm{IO}} = \prod_{k'} P_{\ell k'}^{\omega_{k'k}}$, with spending on sector k' equal to the share $\omega_{k'k}$ of bundle spending. Minimizing $w_\ell l + P_{\ell k}^{\mathrm{IO}} B$ for one unit of output then gives the unit cost

$$a_i = \frac{\gamma_{h_i \ell}}{\varphi_i} \, w_\ell^{1-\nu_k} (P_{\ell k}^{\mathrm{IO}})^{\nu_k}, \qquad P_{\ell k}^{\mathrm{IO}} = \prod_{k' \in \mathcal{K}} P_{\ell k'}^{\omega_{k'k}}, \tag{4}$$

which is the cost equation of the model section. By Shephard's lemma, for every dollar of production cost the plant pays $1 - \nu_k$ to labor in ℓ and $\nu_k \omega_{k'k}$ for the composite of sector k' in ℓ.

Delivered cost. To sell $q_{i,n}$ units in n the plant produces $d_{\ell n} q_{i,n}$ units. Its variable cost of serving n, exclusive of tariffs, is

$$V_{i,n} = a_i d_{\ell n} q_{i,n}, \tag{5}$$

country n collects $t_{\ell nk} V_{i,n}$, and the delivered marginal cost is

$$c_{i,n} = a_i \, d_{\ell n} \, (1 + t_{\ell nk}). \tag{6}$$

Where the money goes. Out of the variable cost $V_{i,n}$, labor in ℓ receives $(1 - \nu_k)V_{i,n}$ and the market (ℓ, k') receives $\nu_k \omega_{k'k} V_{i,n}$ as intermediate purchases. If the plant serves n, it also pays the fixed cost

$$F_{i,n} = f_k \, w_\ell \, f^d_{\ell n} \tag{7}$$

to labor in ℓ. These are the only payments a plant makes; what remains of its revenue is tariff revenue and profit (Step 3).

3 Step 3: One Market with Given Plants

Fix a market (n, k) and the set of active plants $\mathcal{A}$ in it, with $T_g \subseteq \mathcal{A}$ the plants of group g. Drop the market's indices: σ, E, P, c_i, s_i, $t_i \equiv t_{\ell(i)nk}$. The inputs of this step are the delivered costs c_i and the spending level E. Its outputs are prices, quantities, shares, markups and profits.

3.1 Inverse demand

Let $\rho = (\sigma - 1)/\sigma$ and $Q = (\sum_i q_i^\rho)^{1/\rho}$, the sectoral composite. Since $PQ = E$, the lower-tier rule (1) can be inverted as $p_i = P(q_i/Q)^{-1/\sigma} = (E/Q)(q_i/Q)^{-1/\sigma}$, so that

$$p_i = E \, Q^{-\rho} q_i^{\rho - 1}, \qquad s_i = \frac{p_i q_i}{E} = \frac{q_i^\rho}{Q^\rho}. \tag{8}$$

With $\eta = 1$, E is fixed from the point of view of the groups in the market.

3.2 The group's first-order condition and the markup

Group g chooses q_i for every $i \in T_g$ to maximize $\pi_g = \sum_{i \in T_g} (p_i - c_i) q_i$. Its revenue is

$$R_g = \sum_{j \in T_g} p_j q_j = E \, Q^{-\rho} \sum_{j \in T_g} q_j^\rho = E \, S_g, \qquad S_g \equiv \sum_{j \in T_g} s_j.$$

Using $\partial \ln Q / \partial q_i = q_i^{\rho - 1} / Q^\rho = s_i / q_i$, the marginal revenue of plant i is

$$\frac{\partial R_g}{\partial q_i} = E \, Q^{-\rho} \Big[\rho \, q_i^{\rho - 1} - \rho \, \frac{q_i^{\rho - 1}}{Q^\rho} \sum_{j \in T_g} q_j^\rho \Big] = \rho \, p_i \, (1 - S_g).$$

The second term is cannibalization: expanding plant i lowers the price index and takes sales from every variety in the market, including the group's other plants, and the group counts the loss on all of them, which is proportional to its total share S_g. Setting marginal revenue equal to marginal cost c_i gives

$$p_i = \mu(S_g) \, c_i, \qquad \mu(S) = \frac{\sigma}{(\sigma - 1)(1 - S)}. \tag{9}$$

This is Proposition 1 of the model section at $\eta = 1$. The markup is common to all plants of the group and depends only on its total share. It rises from $\mu(0) = \sigma/(\sigma-1)$ and is unbounded as $S \to 1$, which is why every market needs at least two active groups. The markup elasticity, used below, is

$$\varepsilon_\mu(S) \equiv \frac{S\mu'(S)}{\mu(S)} = \frac{S}{1-S}. \tag{10}$$

3.3 Shares within a group

With a common markup, $p_i/p_j = c_i/c_j$ for any two plants of the same group, so by (3) $s_i/s_j = (c_i/c_j)^{1-\sigma}$. Summing over the group,

$$\frac{s_i}{S_g} = \frac{c_i^{1-\sigma}}{K_g}, \qquad K_g \equiv \sum_{i \in T_g} c_i^{1-\sigma}. \tag{11}$$

Costs show up in how the group splits its sales across plants, not in markups.

3.4 The share equation

Let $A \equiv \sum_{i \in \mathcal{A}} p_i^{1-\sigma} = P^{1-\sigma}$. Then $s_i = p_i^{1-\sigma}/A$. Substituting $p_i = \mu(S_g)c_i$ and summing over the group,

$$S_g = \sum_{i \in T_g} \frac{\mu(S_g)^{1-\sigma} c_i^{1-\sigma}}{A} = \frac{\mu(S_g)^{1-\sigma} K_g}{A},$$

that is,

$$\psi(S_g) = \frac{K_g}{A}, \qquad \psi(S) \equiv S\,\mu(S)^{\sigma-1} = \Big(\frac{\sigma}{\sigma-1}\Big)^{\sigma-1} \frac{S}{(1-S)^{\sigma-1}}. \tag{12}$$

For a given A this is one equation in one unknown for each group. Taking logs and moving the constant to the right,

$$\ln S_g - (\sigma-1)\ln(1-S_g) = \ln K_g - \ln A - (\sigma-1)\ln\frac{\sigma}{\sigma-1} \equiv x_g(A). \tag{13}$$

The left side rises strictly from $-\infty$ to $+\infty$ as S_g goes from 0 to 1, so the equation has exactly one root $S_g(A) \in (0,1)$ for every $A > 0$. It has no closed form except in special cases, but it is a monotone scalar equation and Newton's method converges quickly from any interior start, using the derivative

$$\frac{d}{dS}\big[\ln S - (\sigma-1)\ln(1-S)\big] = \frac{1+(\sigma-2)S}{S(1-S)}.$$

3.5 The market-clearing equation

Shares must add up to one. With $S_g(A)$ the root of (13), the equilibrium of the market is the value of A that solves

$$\mathcal{S}(A) \equiv \sum_g S_g(A) = 1. \tag{14}$$

Differentiating (12) in logs, $[1 + (\sigma-1)\varepsilon_\mu(S_g)]\, d\ln S_g = -d\ln A$, so

$$\frac{dS_g}{d\ln A} = -S_g\Lambda_g, \qquad \Lambda_g \equiv \frac{1}{1+(\sigma-1)\varepsilon_\mu(S_g)} = \frac{1-S_g}{1+(\sigma-2)S_g} > 0. \tag{15}$$

Every share falls as A rises, so $\mathcal{S}$ is strictly decreasing and (14) has at most one solution (Proposition 2). A bracket that contains it is found as follows. Since $\mu(S) \geq \mu(0)$, (12) gives $S_g \leq K_g/[A\,\mu(0)^{\sigma-1}]$, so at

$$A_{\text{hi}} = \mu(0)^{1-\sigma} \sum_g K_g$$

shares add up to at most one. As $A \to 0$ every root of (13) tends to one, so $\mathcal{S}$ tends to the number of groups, which is at least two; halving A from A_{hi} reaches a point A_{lo} with $\mathcal{S}(A_{\text{lo}}) \geq 1$ in finitely many steps. Between the two, Newton's method on $\ln A$ with derivative $\mathcal{S}'(\ln A) = -\sum_g S_g \Lambda_g$, falling back to bisection whenever a step would leave the bracket, converges to the unique root.

3.6 Recovering everything else

Once A and the shares S_g are known, every other object in the market follows in closed form:

$$\mu_g = \mu(S_g), \qquad p_i = \mu_g c_i, \qquad P = A^{1/(1-\sigma)}, \tag{16}$$

$$s_i = S_g \,\frac{c_i^{1-\sigma}}{K_g}, \qquad r_i = s_i E, \qquad q_i = \frac{r_i}{p_i}, \tag{17}$$

$$V_i = \frac{r_i}{\mu_g(1+t_i)}, \qquad \text{tariff revenue } t_i V_i. \tag{18}$$

The price index is consistent with the prices, since $\sum_i p_i^{1-\sigma} = \sum_g \mu_g^{1-\sigma} K_g = A \sum_g S_g = A$. Variable profit is

$$\pi_g = \sum_{i \in T_g} (p_i - c_i) q_i = \Big(1 - \frac{1}{\mu_g}\Big) \sum_{i \in T_g} r_i = E\, S_g\, m(S_g), \qquad m(S) \equiv 1 - \frac{1}{\mu(S)} = \frac{1 + (\sigma-1)S}{\sigma},$$

so that

$$\pi_g = \frac{E}{\sigma}(S_g + \kappa S_g^2), \qquad \kappa = \sigma - 1. \tag{19}$$

Profit does not depend on the tariff directly: the tariff enters through c_i, which moves shares.

The revenue identity. For each plant, revenue splits exactly into variable cost, tariff revenue and profit:

$$r_i = \mu_g(1+t_i)V_i = V_i + t_i V_i + (\mu_g - 1)(1+t_i)V_i, \tag{20}$$

and $(\mu_g - 1)(1+t_i)V_i = (p_i - c_i)q_i$ is plant i's contribution to π_g. Per dollar of spending in the market, the three parts are

$$\underbrace{\sum_i \frac{s_i}{\mu_{g(i)}(1+t_i)}}_{\text{variable cost}} + \underbrace{\sum_i \frac{t_i s_i}{\mu_{g(i)}(1+t_i)}}_{\text{tariffs}} + \underbrace{\sum_g S_g\, m(S_g)}_{\text{profit}} = 1. \tag{21}$$

Two properties used below. First, shares, markups and the price index depend only on the delivered costs, never on E: spending scales quantities, revenue, costs and profits, nothing else. Second, multiplying every delivered cost by λ multiplies every K_g and A by $\lambda^{1-\sigma}$, leaves every share and markup unchanged, and multiplies every price and P by λ. The price index is therefore homogeneous of degree one in the delivered costs.

Under price competition. Demand gives $\partial \ln q_i / \partial \ln p_j = -\sigma \mathbf{1}\{i = j\} + (\sigma - \eta) s_j$. Imposing a common markup in the group's first-order conditions gives $\mu(S) = \varepsilon(S)/[\varepsilon(S) - 1]$ with $\varepsilon(S) = \sigma - (\sigma - \eta)S$ (Proposition 1). Steps 3.3–3.6 are unchanged with this μ: the within-group rule (11), the share equation $\psi(S_g) = K_g/A$ with $\psi(S) = S\mu(S)^{\sigma-1}$, the clearing equation and the bracket all carry over, and $\Lambda_g = 1/[1 + (\sigma - 1)\varepsilon_\mu(S_g)]$ with the new ε_μ. The Lerner index is $1/\varepsilon(S)$, so $\pi_g = ES_g/\varepsilon(S_g)$, which is not quadratic in S_g.

When $\eta > 1$ (quantity competition). The markup is $1/\mu(S) = 1 - (1 - S)/\sigma - S/\eta$, which is bounded above by $\eta/(\eta - 1)$, and $\kappa = \sigma/\eta - 1$. Because $\psi(1) = \mu(1)^{\sigma-1}$ is finite, the share equation has an interior root only when $K_g/A < \psi(1)$; the clearing equation is solved on $A > \max_g K_g/\psi(1)$, where $\mathcal{S}$ is still strictly decreasing.

4 Step 4: Entry into One Market

Now let each group choose which of its potential plants serve the market. The inputs are the delivered costs c_i and fixed costs F_i of all potential plants $i \in \mathcal{P}_g$ and the spending level E. The outputs are the active sets T_g and the aggregate A.

4.1 The value of a plant set

By Step 3, a set $T \subseteq \mathcal{P}_g$ affects the group's profit only through its strength $K(T) = \sum_{i \in T} c_i^{1-\sigma}$, and costs it $\Phi(T) = \sum_{i \in T} F_i$. Taking A as given, the group's share with set T is $\psi^{-1}(K(T)/A)$, and its net payoff is

$$U_g(T; A) = \Omega\big(K(T); A\big) - \Phi(T), \qquad \Omega(K; A) = \frac{E}{\sigma}\big(S + (\sigma - 1)S^2\big)\Big|_{S = \psi^{-1}(K/A)}. \tag{22}$$

The empty set gives $U_g = 0$.

4.2 The ladder

A group with $|\mathcal{P}_g|$ potential plants has $2^{|\mathcal{P}_g|}$ sets. Since Ω is strictly increasing in K, a set can be chosen only if no other set has at least as much strength at no greater fixed cost. The construction is:

1. compute $(K(T), \Phi(T))$ for every $T \subseteq \mathcal{P}_g$;
2. discard every T for which some T' has $K(T') \geq K(T)$ and $\Phi(T') \leq \Phi(T)$, with one inequality strict;
3. order the remaining sets by strength. Fixed cost rises along the order too, otherwise a set would have been discarded. Call them rungs $0, 1, \ldots, R_g$, with rung 0 the empty set.

When all of a group's plants pay the same fixed cost F in the market, the ladder is explicit: rung r is the set of the r plants with the lowest delivered costs, since among sets with r plants, which all cost rF, these have the most strength. A stand-alone firm has a single plant and a ladder with two rungs, out or in.

4.3 The best response at a given aggregate

At a given A, the group's best response is the rung with the highest payoff,

$$T_g^*(A) = \arg\max_{r=0,\ldots,R_g} \Omega(K_r; A) - \Phi_r, \qquad K_g^*(A) = K\big(T_g^*(A)\big). \tag{23}$$

In general this is a comparison of R_g+1 numbers. In the equal-fixed-cost case it reduces to a cutoff. The strength added by each rung, $c_{(r+1)}^{1-\sigma}$, falls with r because plants are added from cheapest to most expensive, and Ω is strictly concave in K (Theorem 1(a)). So the gain from climbing one more rung, $\Omega(K_{r+1}; A) - \Omega(K_r; A) - F$, falls with r, and the group climbs while this gain is positive.

4.4 The entry equilibrium of the market

An entry equilibrium (Definition 1) is a pair $(\{T_g\}, A)$ in which every T_g is a best response at A and A clears the market with the chosen strengths. Substituting the best responses into (14) reduces it to one equation in A:

$$\mathcal{S}^*(A) \equiv \sum_g \psi^{-1}\Big(\frac{K_g^*(A)}{A}\Big) = 1. \tag{24}$$

When no group would hold more than half of the market with any of its rungs at the aggregates compared, each $K_g^*(A)$ is non-increasing in A (Corollary 1): a more competitive market makes each group climb fewer rungs. Each term of $\mathcal{S}^*$ then falls with A for two reasons, a lower strength and a higher aggregate, so $\mathcal{S}^*$ is decreasing and crosses one at most once. It is solved by bisection on $\ln A$, with the bracket of Step 3 computed from the strengths of the full sets $\mathcal{P}_g$ at the top and halving until $\mathcal{S}^* \geq 1$ at the bottom, which is possible when at least two groups can cover their fixed costs, the case the model section considers.

The integer problem. Because $K_g^*(A)$ moves in jumps, $\mathcal{S}^*$ can jump over one. The bisection then converges to the aggregate $\bar{A}$ at which the jump occurs, with $\mathcal{S}^*$ above one just below $\bar{A}$ and below one just above it. At $\bar{A}$ some group switches rungs. If it is assigned its larger rung, the aggregate that clears the market lies at or above $\bar{A}$, where the smaller rung is its best response; if it is assigned its smaller rung, the clearing aggregate lies below $\bar{A}$, where the larger rung is. Either way the group wishes to deviate, and no exact equilibrium in whole plants exists. The procedure keeps the configuration with the fewest groups wishing to deviate and reports how many there are.

Scale. Multiplying every delivered cost, fixed cost and spending level by the same λ multiplies every payoff (22) by λ and leaves shares unchanged, so it leaves every best response and the entry equilibrium unchanged. In general equilibrium, this is why entry is unaffected by a common change in all wages.

5 Step 5: Input Prices, Given Wages and Active Sets

From now on, wages w and active sets T are given. The delivered cost of every plant depends on input prices through (4), and input prices are the price indices that those costs generate through Step 3. Substituting one into the other gives a fixed point in the price indices.

The map. Start from a vector of price indices $P = \{P_{\ell k}\}$ and compute, in order:

1. the input bundle price $P^{\mathrm{IO}}_{\ell k} = \prod_{k'} P^{\omega_{k'k}}_{\ell k'}$ in every country and sector;
2. the unit cost a_i of every plant, by (4), and its delivered cost $c_{i,n}$ in every market it serves, by (6);
3. in every market, the strengths K_g, the root A of (14) and the new price index $A^{1/(1-\sigma)}$, by Step 3.

Call the result $\mathcal{T}(P; w)$. Input prices are a fixed point, $P = \mathcal{T}(P; w)$.

Why the fixed point is unique. In logarithms, raising every price index by $x \geq 0$ raises $\ln P^{\mathrm{IO}}_{\ell k}$ by x, $\ln a_i$ by $\nu_k x$, and $\ln c_{i,n}$ by $\nu_k x$. Since each market's price index is homogeneous of degree one and non-decreasing in its delivered costs (Step 3), the new log price index of a sector-k market rises by at most $\nu_k x$. Hence

$$\ln \mathcal{T}(\ln P + x\mathbf{1}) \leq \ln \mathcal{T}(\ln P) + (\max_k \nu_k)\, x\mathbf{1},$$

and the map is monotone. By Blackwell's sufficient conditions it is a contraction in the sup norm with modulus $\max_k \nu_k < 1$ (Proposition 3(a)). It has exactly one fixed point, and iterating from any start converges to it: after j iterations the error is at most $(\max_k \nu_k)^j/(1 - \max_k \nu_k)$ times the size of the first update.

Without intermediates. When $\nu_k = 0$ for all k, the unit cost is $a_i = \gamma_{h_i \ell} w_\ell / \varphi_i$, delivered costs are known directly from wages, and this step is a single pass through Step 3.

Output. For given (w, T), this step delivers every delivered cost $c_{i,n}$, every share $s_{i,n}$ and $S_{g,n}$, every markup $\mu_{g,n}$ and every price index P_{nk}. None of them depends on market sizes or incomes. Scaling all wages by λ scales every price index and every cost by λ and leaves all shares and markups unchanged.

6 Step 6: Market Sizes and Incomes

With prices known, everything that remains is linear in spending. This step substitutes Step 3 into the spending and income equations and solves the resulting linear system.

6.1 Spending as a function of incomes

By (18), the variable cost of plant i in market (n', k') is a known fraction of that market's spending:

$$V_{i,n'} = \frac{s_{i,n'}}{\mu_{g(i),n'}(1 + t_{\ell(i)n'k'})}\, E_{n'k'}.$$

Collect, for each country n, the part of market (n', k')'s spending that becomes variable cost of plants located in n:

$$v_{n,(n',k')} \equiv \sum_{i \in \mathcal{A}_{n'k'}:\, \ell(i) = n} \frac{s_{i,n'}}{\mu_{g(i),n'}(1 + t_{nn'k'})}. \tag{25}$$

Plants of sector k' located in n spend the share $\nu_{k'}\omega_{kk'}$ of their variable costs on sector k in n (Step 2). Spending on market (n, k) is final spending (2) plus these purchases:

$$E_{nk} = \beta_k X_n + \sum_{k' \in \mathcal{K}} \omega_{kk'} \nu_{k'} \sum_{n' \in \mathcal{N}} v_{n,(n',k')}\, E_{n'k'}. \tag{26}$$

Stack the NK market sizes in a vector E and the N incomes in X. Define the $NK \times N$ matrix $\mathcal{E}$ with entries $\mathcal{E}_{(n,k),j} = \beta_k \mathbf{1}\{j = n\}$ and the $NK \times NK$ matrix B with entries $B_{(n,k),(n',k')} = \omega_{kk'}\nu_{k'}\upsilon_{n,(n',k')}$. Then (26) reads

$$E = \mathcal{E}X + BE \quad \Longrightarrow \quad E = (I - B)^{-1}\mathcal{E}\, X \equiv G\, X. \tag{27}$$

The inverse exists and is non-negative. The column of B for market (n', k') sums to

$$\sum_{n,k} \omega_{kk'}\nu_{k'}\upsilon_{n,(n',k')} = \nu_{k'} \sum_{i \in \mathcal{A}_{n'k'}} \frac{s_{i,n'}}{\mu_{g(i),n'}(1 + t_{\ell(i)n'k'})} < \nu_{k'} < 1,$$

using $\sum_k \omega_{kk'} = 1$ and the fact that markups exceed one and tariffs are non-negative. A non-negative matrix whose columns sum to less than one has spectral radius below one, so $(I - B)^{-1} = \sum_{s \geq 0} B^s \geq 0$. The entry $G_{(n,k),j}$ is the spending on market (n, k) generated, directly and through all rounds of intermediate purchases, by one dollar of final spending by country j.

6.2 Incomes as a function of spending

Income is labor income plus owned net profits plus tariff revenue:

$$X_n = w_n L_n + \sum_g \theta_{gn}\Pi_g + T_n.$$

Write each term in terms of spending. Group g of sector k earns variable profit $E_{n'k}S_{g,n'}m(S_{g,n'})$ in each destination n' by (19), and pays the fixed costs of its active plants,

$$\Phi_g \equiv \sum_{n'} \sum_{i \in T_{g,n'}} f_k\, w_{\ell(i)}\, f^d_{\ell(i)n'}, \qquad \Pi_g = \sum_{n'} E_{n'k}S_{g,n'}m(S_{g,n'}) - \Phi_g.$$

The fixed costs depend on wages and active sets only. Tariff revenue in n is, by (18),

$$T_n = \sum_k \tau_{nk}E_{nk}, \qquad \tau_{nk} \equiv \sum_{i \in \mathcal{A}_{nk}} \frac{t_{\ell(i)nk}\, s_{i,n}}{\mu_{g(i),n}(1 + t_{\ell(i)nk})}.$$

Define the $N \times NK$ matrix H with entries

$$H_{n,(n',k)} = \sum_{g:\, k(g)=k} \theta_{gn}S_{g,n'}m(S_{g,n'}) + \mathbf{1}\{n' = n\}\, \tau_{nk}, \tag{28}$$

the share of market (n', k)'s spending that becomes income of country n as profit or tariff revenue. Then

$$X = wL - \theta'\Phi + H\, E.$$

6.3 Solving the system

Substituting (27),

$$X = wL - \theta'\Phi + \mathcal{M}X, \qquad \mathcal{M} \equiv HG, \qquad \Longrightarrow \qquad X = (I - \mathcal{M})^{-1}(wL - \theta'\Phi), \tag{29}$$

and then $E = GX$. Here wL is the vector with entries w_nL_n and $\theta'\Phi$ the vector with entries $\sum_g \theta_{gn}\Phi_g$.

The inverse exists because every column of $\mathcal{M}$ sums to less than one. Consider one dollar of final spending by country j. It generates the spending Ge_j across markets. By (21), the part of each market's spending that is neither profit nor tariff revenue is variable cost. So the profits and tariffs generated, which is the column sum $\mathbf{1}'\mathcal{M}e_j$, equal total spending generated minus total variable cost generated. Total spending generated is the one dollar of final spending plus all intermediate purchases, and intermediate purchases are the share ν of variable costs. Hence

$$\mathbf{1}'\mathcal{M}e_j = 1 - \sum_i (1-\nu_{k(i)}) \left[\text{variable cost of plant } i \text{ generated by the dollar}\right] < 1,$$

since the generated variable wage bill is positive when $\nu_k < 1$. With $\mathcal{M} \geq 0$ and column sums below one, $\rho(\mathcal{M}) < 1$ and the solution is unique (Proposition 3(b)).

Without intermediates. With $\nu_k = 0$, $B = 0$, $G = \mathcal{E}$ and $E_{nk} = \beta_k X_n$. The income system becomes N equations,

$$X_n = w_n L_n - \sum_g \theta_{gn}\Phi_g + \sum_{n'}\Big[\sum_g \theta_{gn}\beta_{k(g)}S_{g,n'}m(S_{g,n'})\Big]X_{n'} + \Big[\sum_k \beta_k \tau_{nk}\Big]X_n.$$

Output. For given (w, T), Steps 5 and 6 deliver every market size E_{nk}, every income X_n, every group's net profit Π_g and every country's tariff revenue T_n, and through (17)–(18) every quantity and variable cost. Scaling all wages by λ scales the right side of (29) by λ and leaves $\mathcal{M}$ unchanged, so incomes and market sizes scale by λ too.

7 Step 7: The Labor Market

Labor demand. Country n's wage bill is what plants located in n pay to labor: the share $1-\nu_k$ of their variable costs, plus the fixed costs of the markets they serve. Using (25),

$$W_n(w;T) = \sum_k (1-\nu_k)\sum_{n'} v_{n,(n',k)}E_{n'k} + \sum_{n'}\sum_{i:\,\ell(i)=n,\ i\in T_{g(i),n'}} f_{k(i)}\, w_n\, f^d_{nn'}. \tag{30}$$

Every term is known from Steps 5 and 6. The excess demand for labor is

$$z_n(w;T) = \frac{W_n(w;T)}{w_n} - L_n, \tag{31}$$

and labor markets clear when $z_n = 0$ for every n.

Walras' law. One of these conditions is redundant: at any wages,

$$\sum_n w_n z_n(w;T) = \sum_n W_n - \sum_n w_n L_n = 0.$$

To see it, add the revenue identity (20) over all plants and markets:

$$\sum_{\text{markets}} E_{nk} = \sum V + \sum_n T_n + \sum_g \pi_g,$$

where $\sum V$ is total variable cost and π_g is the group's total variable profit. Total spending is final spending, $\sum_n X_n$, plus intermediate purchases, $\sum \nu V$. So $\sum_n X_n = \sum(1-\nu)V + \sum_n T_n + \sum_g \pi_g$.

Adding the income equations over countries, with $\sum_n \theta_{gn} = 1$, $\sum_n X_n = \sum_n w_n L_n + \sum_g (\pi_g - \Phi_g) + \sum_n T_n$. Equating the two expressions,

$$\sum_n w_n L_n = \sum (1-\nu) V + \sum_g \Phi_g = \sum_n W_n,$$

since every fixed cost is paid to labor in the plant's location.

The numeraire. Scaling all wages by λ scales costs, prices and fixed costs by λ (Step 5), incomes and market sizes by λ (Step 6), and leaves shares, markups and active sets unchanged (Step 4). By (30), W_n scales by λ, so z is homogeneous of degree zero. Only relative wages are determined, and we set $w_1 = 1$. By Walras' law, $z_1 = 0$ whenever $z_2 = \cdots = z_N = 0$.

8 Step 8: Wages, Given Active Sets

The reduced system. Steps 5–7 turn the whole model, for given active sets, into $N-1$ equations in $N-1$ unknowns:

$$z_n(1, w_2, \ldots, w_N; T) = 0, \qquad n = 2, \ldots, N. \tag{32}$$

Evaluating $z(w;T)$ at a given w means: solve the price fixed point (Step 5), build B, H and Φ and solve the linear system (Step 6), and compute the wage bills (30).

Fully written out without intermediates. When $\nu_k = 0$ the reduced system can be written explicitly in terms of wages and market shares. Delivered costs are $c_{i,n} = \gamma_{h_i \ell} w_\ell d_{\ell n}(1 + t_{\ell n k})/\varphi_i$, so shares in every market depend on wages only through relative delivered costs. Spending is $E_{nk} = \beta_k X_n$, and incomes solve the N-equation system of Step 6. The wage bill is

$$W_n = \sum_{n'} \sum_k \beta_k X_{n'} \sum_{i \in \mathcal{A}_{n'k}:\, \ell(i) = n} \frac{s_{i,n'}}{\mu_{g(i),n'}(1 + t_{nn'k})} + w_n \sum_{n'} \sum_{i:\, \ell(i) = n,\ i \in T_{g(i),n'}} f_{k(i)} f^d_{nn'},$$

and (32) requires $W_n = w_n L_n$. A rise in a foreign wage w_j affects W_n in three ways: through the shares, as plants located in n gain sales from plants located in j; through the markups that the share changes imply; and through incomes $X_{n'}$, which include the profits of the groups each country owns. The last channel is the ownership channel of the model.

Damped tâtonnement. The solution algorithm updates wages by

$$w_n^{(j+1)} = \big(w_n^{(j)}\big)^{1-\lambda} \Big(\frac{W_n(w^{(j)};T)}{L_n}\Big)^{\lambda}, \qquad \text{that is,} \qquad \ln w_n^{(j+1)} = \ln w_n^{(j)} + \lambda \ln\Big(1 + \frac{z_n(w^{(j)};T)}{L_n}\Big), \tag{33}$$

and then rescales so that $w_1 = 1$, which is allowed because the map is homogeneous of degree one. A country's wage rises when its labor is in excess demand. A wage vector is a fixed point exactly when $W_n = w_n L_n$ for every n, so the fixed points are the equilibrium wages.

How much damping. Let $M_{nj} = \partial \ln W_n / \partial \ln w_j$. In logarithms the map (33) has Jacobian $(1-\lambda)I + \lambda M$, whose rows sum to one because W is homogeneous of degree one. Its off-diagonal entries are positive when labor is a gross substitute, $M_{nj} > 0$ for $n \neq j$. Its diagonal entries are positive when

$$\lambda < \frac{1}{1 - \min_n M_{nn}}.$$

When labor demand is elastic, $M_{nn} < 0$, the bound is below one, and an undamped update ($\lambda = 1$) can overshoot. When both conditions hold on a compact set of wages that is convex in logarithms, the map is a contraction in Hilbert's projective metric on that set (Theorem 2), and the iteration converges to the unique equilibrium in it.

Newton steps. Close to the solution, Newton's method on $F_n(\ln w) = \ln W_n - \ln w_n - \ln L_n$, $n = 2, \ldots, N$, converges faster. Its Jacobian is

$$\frac{\partial F_n}{\partial \ln w_j} = M_{nj} - \mathbf{1}\{n = j\},$$

restricted to $n, j \geq 2$. The matrix M can be obtained column by column, by perturbing one $\ln w_j$ and re-evaluating Steps 5–7. The step is $\Delta \ln w = -[M - I]^{-1}_{-1,-1} F$. Iterations stop when $\max_n |z_n|/L_n$ falls below a tolerance.

9 Step 9: Active Sets

Steps 5–8 hold the active sets fixed. In equilibrium, the active sets in every market must form an entry equilibrium (Step 4) at the delivered costs, fixed costs and market sizes that the wages generate. These depend on the active sets in all markets: through input prices, which affect costs everywhere, and through incomes, which affect spending everywhere. Active sets are therefore found by an outer loop.

1. Start from a configuration $T^{(0)}$, for example every potential plant active in every market of its sector.
2. Given $T^{(j)}$, solve Steps 5–8 for wages, and record in every market the delivered costs $c_{i,n}$, the fixed costs $F_{i,n}$, the market size E_{nk} and the aggregate A_{nk}.
3. In every market, compute each group's ladder and its best response at the recorded aggregate (Steps 4.2–4.3), and its gain $\Delta_{g,n} = U_g(T^*_{g,n}; A_{nk}) - U_g(T^{(j)}_{g,n}; A_{nk}) \geq 0$.
4. If no group gains, stop: every group's set is a best response at its market's aggregate, and the aggregate clears the market with those sets, so every market is in an entry equilibrium and $(w, T^{(j)})$ is an equilibrium.
5. Otherwise, let the groups with the largest gains switch to their best responses, call the result $T^{(j+1)}$, and return to step 2.

Moving only the groups with the largest gains is a way of damping the outer loop. Moving every group at once can make the configuration oscillate, because wages computed for a dense configuration differ substantially from wages for a sparse one. The choice affects the path of the algorithm, not its fixed points, and it does not impose an order of entry. The loop is not guaranteed to converge. When it does not reach a configuration in which no group wishes to move, it returns the one with the fewest such groups and reports how many there are. Because the joint equilibrium of wages and active sets is not known to be unique, the whole procedure is run from several starting points and the resulting configurations are compared.

10 The Complete Algorithm, and What to Check

Putting the steps together, from the innermost loop outward:

(A) *One market, plants given (Step 3).* Given delivered costs: solve each share equation (13) for $S_g(A)$, and the clearing equation (14) for A; recover μ_g, p_i, P and s_i by (16)–(17).

(B) *Input prices (Step 5).* Given (w, T): iterate $P \mapsto \mathcal{T}(P; w)$, which calls (A) in every market, until it converges.

(C) *Spending and incomes (Step 6).* Given (w, T) and the output of (B): build B, H and Φ; compute X by (29) and $E = GX$.

(D) *Labor market (Step 7).* Compute W_n by (30) and z_n by (31).

(E) *Wages (Step 8).* Update w by (33), then by Newton steps, repeating (B)–(D), until $z(w; T) = 0$.

(F) *Active sets (Step 9).* At the wages from (E), compute every group's best response in every market; if some group gains, update the configuration and return to (B).

Closed form	Computed
demand, unit and delivered costs	each group's share, given A (monotone scalar equation)
markups, prices, quantities given shares	each market's A (monotone scalar equation)
within-group shares, variable profits	each group's ladder and best response (finite comparison)
revenue, cost and tariff splits	input prices (contraction)
wage bills given spending and shares	spending and incomes (linear system)
	wages ($N - 1$ equations) and active sets (outer loop)

Checks at the solution. Each of the following holds exactly at an equilibrium and is a useful test of any implementation.

1. In every market, shares add up to one and $\sum_i p_i^{1-\sigma} = P^{1-\sigma}$.
2. For every plant, revenue equals variable cost plus tariff plus profit, (20).
3. Every country's income equals labor income plus owned net profits plus tariff revenue, and world income equals world final spending.
4. Every labor market clears, and Walras' law holds also away from the solution, which tests the accounting at arbitrary wages.
5. No group gains from changing its set at its market's aggregate.
6. The hypotheses of the uniqueness results: no group holds more than half of any market (Theorem 1), and the wage-bill elasticities satisfy the gross-substitutes and damping conditions of Theorem 2.